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OPINION

Priorities in smart grid design research

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Introduction

Smart grids (SGs) are energy distribution grids augmented by digital technology, such as sensors and communication systems. Their goal is to improve the efficiency, reliability, resilience, and sustainability of energy supply [1]. Over the last two decades, SGs have attracted much interest among researchers and policymakers. SGs offer solutions for low voltage grids with increasing installation and interoperability of distributed generation, storage, smart appliances and other assets. The features of distribution grids are determined by economic, social, technological, ecological and regulatory/policy requirements [2]. As a result, the speed of SG adoption and diffusion will inevitably vary.

SG research must move beyond focusing solely on technological solutions and blueprints. Notably, the active participation of smart energy consumers and effective regulatory frameworks are crucial for realizing the benefits of SG [3]. This is particularly important in view of the fact that there is a need for research into integrated energy networks [4], i.e., both the technical coupling of different types of networks (heat, gas, electricity) as well as the integration of the digital twins of different sectors (energy, transport, industry, etc.). SG adoption can be hindered or promoted by economic aspects, regulatory policies and the technological context of specific network segments. The sociocultural milieu and its level of technological maturity also play a significant role in the introduction of SGs. According to Ref. [5], the contributions of social sciences and humanities are often overlooked in discussions about energy strategies, despite of the fact that citizen engagement and behavioral change are critical for achieving decarbonization and sustainable energy transition goals. Consequently, a comprehensive research agenda geared towards the intersection of Systems&Technology, Economics and Society can help to narrow down important SG research gaps.

Systems and technology

Although attempts have been made to standardize SG technologies, there is still no universally accepted standard. The technological challenges faced by SGs are similar across different applications, but the technological environment can vary significantly, from differences in grid capacities to non-standardized communication and control protocols. Smaller grid capacities require more optimization for load balancing, and as smart assets in grids become more numerous, the amount of data collected will rapidly increase, necessitating efficient data science solutions. To address these challenges, standardization efforts are needed, along with the integration of cloud computing technologies and SG reconfiguration techniques, such as the cloud-based architecture for the automation and management of SGs [6]. Such architectures can improve scalability, interoperability, and flexibility while integrating a range of grid automation services, ultimately enabling more efficient and sustainable energy management.

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As described above, integration of different systems at the data level is also necessary. Systems can be mapped as a digital twin with the help of field data. Such sectoral digital twins (transport, industry, energy, etc.) can form the bridge for an integrated energy system. Although models for the individual systems are already being researched, we see the need for research on the integration of such digital twins.

Economics

Research on implementation strategies and net gains of different SG technologies is crucial for policy makers. Implementation can be encouraged by introducing competition among distribution system operators (DSOs), incentive-based regulatory schemes, and innovation-stimulus mechanisms. Ref. [7] did a study on 459 innovative SG projects in 30 European countries; Ref. [8] used an economic valuation model to evaluate the net gains of different implementation strategies for SG technologies.

SG risks are another area which warrants a holistic analysis. This means to systematically address technical, financial, market, environmental and other risks. Current literature in the context of SG risk analysis has typically used traditional measures to quantify risk and often dealt with specific SG elements only (e.g., generation, distribution, transmission) rather than treating SG as an integrated system [9].

Regulatory frameworks for DSOs have mostly focused on productivity, including cost- or revenue-based regulation, and more recently, output-/performance-based regulation that can account for the quality of SG supply, and distributed congestion management [1, pp.107-135; 10]. DSOs will need to invest significantly in SG technologies, making it important to understand the sustainability of socio-economic-technological transitions. The multi-level perspective (MLP) approach [11] considers distinct levels–niche (emerging innovation), regime (established practices and dominant technologies) and landscape (socio-political and economic factors). It can be useful in examining the evolution of SG policy and practice. Thereby, the overarching transitions can be better understood [12]. Related to the issue of consumer participation is the need to develop new business models seizing the opportunities enabled by SG. The implementation momentum increases with growing technological bottlenecks, which can be overcome with the help of SGs. This raises the question of how the framework conditions for a more continuous implementation can be designed.

Social acceptance and engagement

SGs have the potential to massively disrupt societies. Hence there is a need (as most research so far focused on technical solutions) to better understand social and customers' acceptance for establishing SGs but also for SG-enabled products and services [13, 14]. There seems to be a wide knowledge gap between "*discourses*, *factual or imaginary argumentation and justifications*, *promises*, *motivations*, *appeals to the public and other narrative elements that scaffold the SG vision in Europe*" and citizen debates identified ([15], p.16).

Conclusions

In conclusion, the question arises which aspects (and from which and whose perspective) of SG transitions should be prioritized by researchers and policy-makers. We have argued that substantial research gaps still exist concerning the design of SGs, not so much in the engineering domains, but especially in data science, decision science, the social sciences and humanities. The reason is the comprehensive energy system transformation alongside a major societal transition. Taking a multi-disciplinary perspective way beyond engineering and ICT R&D, we see a particular need to do more research on standardization, investment needs and

investment decision-making under risk and uncertainty, regulatory and market reform issues, social acceptance, and citizen and consumer engagement. In contrast to disciplinary research, transdisciplinary research efforts are needed as well to raise social acceptance and to come up with more balanced policy recommendations and sustainable energy transition than otherwise.

Author Contributions

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References

1. van Dinther C, Flath C, Madlener R. (Eds.). Smart Grid Economics and Management. 1st ed. Lecture Notes in Energy, 51. Switzerland: Springer Cham; 2022. (ISBN 978-3-030-84284-0, ISBN 978-3-030-84286-4).
2. Ponce-Jara MA, Ruiz E, Gil R, Sancristóbal E, Pérez-Molina C, Castro M. Smart Grid: Assessment of the past and present in developed and developing countries. Energy Strategy Reviews. Elsevier; 2017; 18: 38–52. https://doi.org/10.1016/j.esr.2017.09.011
3. Goulden M, Bedwell B, Rennick-Egglestone S, Rodden T, Spence A. Smart grids, smart users? The role of the user in demand side management. Energy Research & Social Science. 2014; 2: 21–29. https://doi.org/10.1016/j.erss.2014.04.008
4. Jamasb T, Llorca M. Energy system integration: economics of a new paradigm. Economics of Energy & Environmental Policy. 2019.p. 7–28.
5. Brückmann G. Berger S, Caviola H, Hahnel UJJ, Piana V,Sahakian M, et al. Towards more impactful energy research: The salient role of social sciences and humanities. PLOS Climate. 2023; 2(2): e0000132. https://doi.org/10.1371/journal.pclm.0000132
6. Pau M, Patti E, Barbierato L, Estebsari A, Pons E, Ponci F, et al. A cloud-based smart metering infrastructure for distribution grid services and automation. Sustainable Energy, Grids and Networks. 2018; 15: 14–25. https://doi.org/10.1016/j.segan.2017.08.001.
7. Cambini C, Meletiou A, Bompard E, Masera A. Market and regulatory factors influencing smart grid investment in Europe: evidence from pilot projects and implications for reform. Utilities Policy. 2016; 40: 36–47.
8. de Oliveira-de Jesus PM, Henggeler Antunes C. Economic valuation of smart grid investments on electricity markets. Sustainable Energy, Grids and Networks. 2018; 16: 70–90. https://doi.org/10.1016/j.segan.2018.05.003
9. Katina PF, Keating CB, Zio E, Gheorghe AV. A criticality-based approach for the analysis of smart grids. Technol Econ Smart Grids Sustain Energy. 2016; 1: 14. https://doi.org/10.1007/s-40866-016-0013-2
10. Iqtiyanillham N, Hasanuzzaman M, Hosenuzzaman M. European smart grid prospects, policies, and challenges. Renewable and Sustainable Energy Reviews. 2017; 67: 776–790. https://doi.org/10.1016/j.rser.2016.09.014
11. Geels FW. "Micro-foundations of the multi-level perspective on socio-technical transitions: Developing a multi-dimensional model of agency through crossovers between social constructivism, evolutionary economics and neo-institutional theory." Technological Forecasting and Social Change. 2020; 152: 119894.
12. Winfield M, Weiler S. Institutional diversity, policy niches, and smart grids: A review of the evolution of Smart Grid policy and practice in Ontario, Canada. Renewable and Sustainable Energy Reviews. 2018; 82: 1931–1938. https://doi.org/10.1016/j.rser.2017.06.014
13. Ellabban O, Abu-Rub H. Smart grid customers' acceptance and engagement: an overview. Renewable and Sustainable Energy Reviews. 2016; 65: 1285–1298.doi.org/10.1016/j.rser.2016.06.21
14. Bigerna S, Bollino CA, Micheli S. Socio-economic acceptability for smart grid development–a comprehensive review. Journal of Cleaner Production. 2016; 131: 399–409. https://doi.org/10.1016/j.jclepro.2015.05.010

15. Vesnic-Alujevic L, Breitegger M, Guimarães Pereira Â. What smart grids tell about innovation narratives in the European Union: Hopes, imaginaries and policy. Energy Research & Social Science. 2016; 12: 16–26. https://doi.org/10.1016/j.erss.2015.11.011